

iTantra

Indian Multilingual TTS &
STT Aided Neural
Transceiver — Radio Access
for Low Bitrate Links

Speech goes in one end. Speech comes out the other. In between it becomes forty-four bytes — small enough to cross a radio link that could never carry a voice.

ORGANISATION	DEPARTMENT	CATEGORY
ISRO	Department of Space	Software
THEME	INSTITUTE	TEAM ID
Smart Automation	Vignana Bharathi Institute of Technology	—
TEAM NAME	REPORT REVISION	
—	1.0	

SECTION 01

Abstract

A disaster radio link carries a few hundred bits per second. Human speech, as audio, needs several thousand. iTantra closes that gap by transmitting meaning instead of waveform.

In alert and distress scenarios, voice is the only interface that includes everyone — literate or not, gloved or not, in any of ten Indian scripts. But the links that survive when infrastructure fails, LoRa and HF and narrowband satellite, cannot carry a voice call. This is not an engineering gap awaiting a better codec; it is a bound. Audio codecs compress the shape of a pressure wave, and a waveform detailed enough to be understood has an irreducible size.

iTantra removes the waveform from the link entirely. An Android application recognises speech on the sending handset, transmits the resulting text as a few dozen bytes, and re-synthesises speech on the receiving handset. Both operators only ever speak and listen; neither reads nor types. Ten Indian languages run entirely on-device, with no cloud, no SIM, no network permission and no proprietary SDK.

2 182×	44 B	<1 s	10	0
COMPRESSION	PER SENTENCE	END TO END	LANGUAGES	NETWORK CALLS

STATUS OF THE FIGURES IN THIS REPORT

The system is fully specified and not yet implemented. Every performance number in this document is a **design target** derived from published characteristics of the selected models and from the frame arithmetic in Section 04 — not a measurement. Section 07 defines exactly how each will be measured, on which device, and under what conditions, before any of them is reported as a result.

SECTION 02

The problem, to scale

Three numbers decide the whole project, and they are far enough apart that a linear axis cannot show them together.

To reproduce a voice with enough fidelity that words remain distinguishable, a microphone samples air pressure sixteen thousand times a second at sixteen bits — a quarter of a million bits for every second of speech. Opus, the best open-source speech codec available, carries an intelligible voice at around six thousand bits per second. A narrowband disaster link offers roughly three hundred.

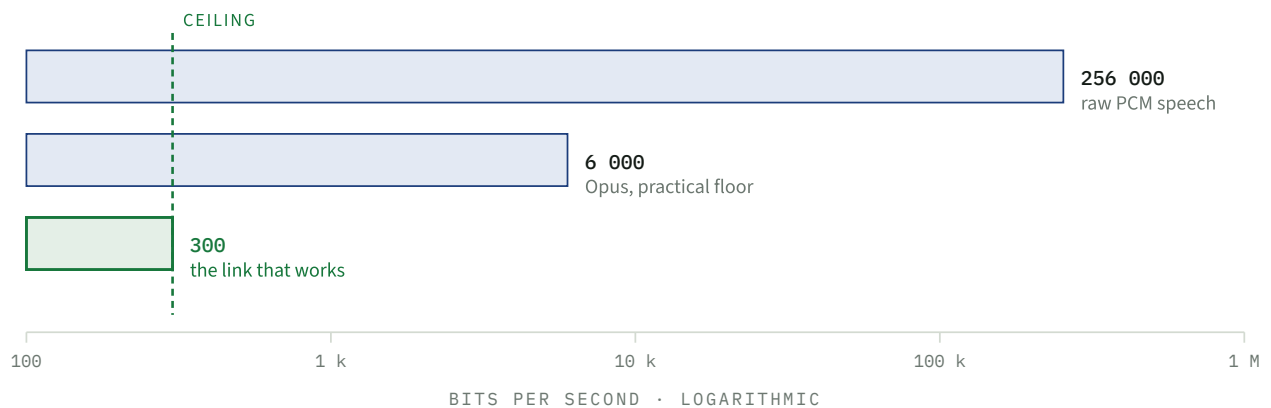


Figure 2.1 — Both audio representations sit far to the right of the link ceiling. The gap between Opus and the link is a factor of twenty, and it cannot be closed by a cleverer codec: it is a property of what a waveform is.

Why the interface must still be voice

If the payload is text, why not let people type it? Because the problem statement's own justification is inclusion. In a disaster the person holding critical information may be a fisherman, a farmer or a village elder; a soldier at altitude in gloves; a rescuer on a rope. Speaking a sentence takes three seconds. Typing it in Devanagari, in the dark, with cold hands, takes forty — if it happens at all.

So the interface is voice, because voice is inclusive. The transmission is text, because text fits. iTantra is the machine between the two.

SECTION 03

Compressing meaning, not waveform

The system degrades gracefully through three levels, choosing the most aggressive one the situation allows.

LEVEL	MECHANISM	3 S HINDI SENTENCE	ON A 300 BPS LINK
—	Raw PCM, 16 kHz 16-bit mono	96 000 B	43 min
—	Opus at 6 kbps	2 250 B	60 s
L1	Semantic transcoding — speech to UTF-8 text	99 B	2.6 s
L2	Script packing — one byte per Indic codepoint	32 B	1.2 s
L3	Template code — one byte per canned sentence	1 B	0.3 s

Level 1 · Semantic transcoding

Speech becomes text. This is where the enormous ratio comes from — roughly 770 to 1 before any further work. Everything below is refinement.

Level 2 · Script packing

UTF-8 spends three bytes per character on Devanagari, Tamil, Kannada and the rest, because it must be able to represent every script at once. But the frame header already declares the language, so the receiver knows which 128-codepoint Unicode block applies. Subtracting the block base yields a single byte per character. The transformation is lossless, table-driven, costs microseconds, and is available only because we control both endpoints.

Real messages are not pure Indic script: they carry Latin digits, punctuation, and the zero-width joiners that shape conjuncts. An escape sequence — 0x1B followed by a 24-bit codepoint — covers anything outside the declared block, at four bytes for a character that would otherwise be silently lost.

Level 3 · Template codes

In an emergency, vocabulary is small and predictable. A shared table maps common operational sentences to single-byte identifiers. After recognition the text is fuzzy-matched against that table; on a confident match the system sends the identifier and the receiver renders the full sentence.

A CONSEQUENCE WORTH NOTICING

Because every handset holds the template table in all ten languages, a template-coded message is spoken in whichever language the *receiver* has selected. A Hindi speaker's alert reaches a Tamil speaker in Tamil — with no translation model, no extra download and no extra latency. **Cross-language operation falls out of the compression scheme.** It applies to template traffic only, and the report claims nothing wider.

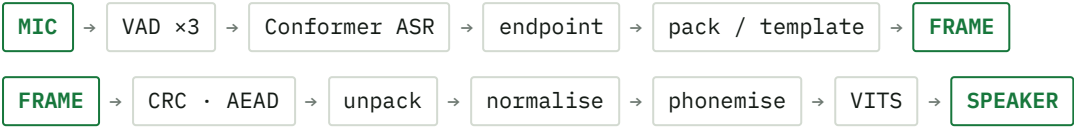
What is discarded, and what compensates

LOST	COMPENSATION
Speaker identity	Sender byte in the header, display name in the interface, and a distinct synthesis voice per node
Emotion and urgency	Explicit priority type; alert traffic routed to the alarm stream with a faster, harder delivery profile
Background audio	Accepted. Out of scope, and arguably desirable given the budget
Recognition errors	Confidence on the wire, recognised text shown before sending, domain vocabulary biasing, confirmation required for alerts

SECTION 04

System architecture

Every device runs the complete system. Role is a runtime mode, not a build variant — one APK, installed on both handsets.



Modules

MODULE	RESPONSIBILITY
app	Compose interface, push-to-talk, alert delivery, provisioning, latency strip, foreground service
core-audio	Capture and playback, ring buffers, alert audio policy, focus handling
core-asr	Three-tier detection, endpointing, recognition, contextual biasing, confidence
core-tts	Normalisation, phonemisation, chunked synthesis, streaming payout
core-link	One <code>Link</code> interface, four transport implementations
core-proto	Frame codec, CRC, AEAD, script packing, template tables, replay window
core-models	Language pack manifest, checksum verification, storage lifecycle
bench	Word error rate harness, latency logger, noise mixing, scorecard export

THE BOUNDARY THAT MATTERS

`core-proto` carries no Android dependency at all. The highest-risk code in the project — the frame codec — is therefore unit-tested and fuzzed on a laptop in seconds rather than on a handset in minutes, and the entire transport layer can be built and proven before a single model exists.

Threading

Audio work never touches the interface thread and the recogniser never blocks capture. Four threads with bounded queues between them: capture at urgent-audio priority, which must not allocate; inference; link; and synthesis. The engine lives in a foreground service, because the normal operating state for a radio is screen-off.

THREAD	DUTY	CONSTRAINT
Capture	20 ms blocks, energy gate, VAD	Zero allocation, no logging
Inference	Encoder and decoder, partial and final hypotheses	May lag transiently; the queue absorbs it
Link	Blocking socket I/O, framing, retries, heartbeat	The only place blocking I/O is expected
Synthesis	Chunked generation feeding <code>AudioTrack</code>	Must stay ahead of the playback clock

SECTION 05

The wire protocol

Ten bytes of header and two of trailer buy channels, priority, relay, integrity and authentication. The payload is what remains.

MAGIC	TYPE	SEQ	FLAGS	LEN	SRC	KEYID	TTL	PAYLOAD	CRC16
0	1	2-3	4	5-6	7	8	9	10 ... n-3	n-2, n-1

Figure 5.1 — Frame layout. Big-endian throughout. There is no destination field: every frame reaches every unit holding the key, exactly as a walkie-talkie does. Selection is by key, not by address.

Frame budget

FRAME	HEADER	PAYLOAD	TAG	TOTAL	300 BPS
Template alert, unauthenticated	12	1	—	13 B	0.3 s
Template alert with coordinates	12	9	8	29 B	0.8 s
Packed Hindi sentence	12	32	—	44 B	1.2 s
Packed sentence, authenticated (LoRa)	12	32	8	52 B	1.4 s
Packed sentence, authenticated (Bluetooth)	12	32	16	60 B	1.6 s
Equivalent Opus audio	—	2 250	—	2 250 B	60 s

REPORTING THE HONEST RATIO

2 182× is the unauthenticated frame against raw PCM. In any deployment worth defending the payload is sealed, and the figure is 1 600×. Against the Opus floor — the comparison a technical reviewer will actually ask for — it is 43×. All four ratios are carried in the evaluation scorecard rather than only the largest.

Integrity and authenticity

The problem statement asks for none of this. It is included because the system wakes a locked, silenced handset and announces at maximum volume on command; a device with that capability and no authentication is a liability rather than a deployable system.

- **AES-256-GCM** with a pre-shared key held by every paired unit. The entire header is associated data, so the sender identity cannot be altered without invalidating the tag.
- **Derived nonce** — EPOCH || SRC || SEQ , never transmitted, saving twelve bytes per frame.
- **Sliding replay window**, 64 entries per sender, so a recorded alert cannot be replayed.
- **Optical key transfer only**. The key crosses by QR code and never touches the radio medium, which is what makes a pre-shared key meaningfully pre-shared.

SECTION 06

Recognition and synthesis

Three-tier detection

Running the acoustic model continuously would exhaust the battery in about two hours and fail the efficiency criterion outright. Detection is tiered, so the expensive stage runs only when the cheap stages agree that a human is speaking.

TIER	MECHANISM	COST / 20 MSPURPOSE
0	Energy gate against an adaptive noise floor	~3 μ sRejects silence — most of elapsed time
1	Silero VAD, 1.8 MB	~300 μ sSeparates speech from wind, engines, doors
2	Conformer acoustic model	~4 msRecognition. Runs only when tier 1 agrees

Endpointing

The problem statement asks for activation "after detecting pauses and stoppages". A state machine finalises on 400 ms of trailing silence in phone mode, or 150 ms after key release in push-to-talk. A 250 ms leading pad is retained so the first phoneme is never clipped, and utterances are bounded at 8 s so one speaker cannot monopolise a half-duplex channel.

The silence window is the single largest latency term — and it is a design choice, not a computational cost. Push-to-talk is measurably faster precisely because releasing the key replaces it.

Contextual biasing

Decoding scores are boosted for a supplied phrase list. In a distress context the critical vocabulary is small and known in advance: मदद, घायल, आग, निकासी, unit callsigns, sector numbers, place names.

This measurably reduces error on precisely the words whose misrecognition would be most costly, and requires no retraining. Negation terms are weighted heavily, because a recognition error that turns "do not evacuate" into "now evacuate" inverts meaning and is the most dangerous failure the system can produce.

Quantisation

Models are trained in 32-bit floating point, which is unnecessary for inference. Dynamic quantisation rewrites the weights as 8-bit integers, exploiting the fact that ARM cores execute integer arithmetic considerably faster than float.

PRECISION	SIZE	RELATIVE SPEED	WER IMPACT	VERDICT
float32, as trained	~120 MB	1.0×	baseline	Unshippable
float16	~60 MB	1.3×	≈ 0	Intermediate
int8 dynamic	~32 MB	2.5×	+1 % rel.	Shipped

Four times smaller and two and a half times faster for roughly one per cent relative degradation. This single transformation is what makes on-device speech viable on an entry-tier handset.

Synthesis

Two of the four synthesis stages are ordinary software engineering rather than machine learning, and they account for most of the perceived quality. **Text normalisation** decides that 112 is dialled digit-wise while 5 km is a quantity — a rule, not a model, and the difference between a synthesiser that sounds professional and one that sounds broken. **Chunked playout** splits the sentence at clause boundaries and begins playing the first chunk while later ones are still generating, cutting time-to-first-audio from roughly 700 ms to 180 ms for the same total work.

SECTION 07

Transport and deployment

Four physical media, one interface. The application never learns which one is carrying its bytes.

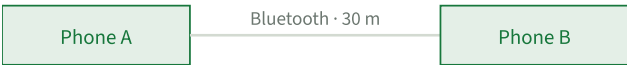
PROPERTY	BT CLASSIC	BLE	WI - FI	LORA SERIAL
Range	10–30 m	10–50 m	50–150 m	2–15 km
Usable rate	~200 kbps	5–20 kbps	>10 Mbps	0.3–5 kbps
Power	Medium	Very low	High	Low
Carries a voice call	Marginally	No	Yes	No
Carries an iTantra sentence	Yes	Yes	Yes	Yes
Role	Default	Standby	Reach	Deployment

Read the last two rows together. On the link that matters operationally — the one that works when infrastructure does not — audio is impossible and iTantra is comfortable. That gap is the product.

The deployment path

Bluetooth RFCOMM is a virtual serial cable, and it is the same profile an ESP32 or a radio modem exposes. *The code that talks to another phone is the code that talks to a radio.* Approximately ₹1 500 of hardware and two hundred lines of firmware convert a phone-to-phone demonstration into a deployable system.

DEMONSTRATION



DEPLOYMENT

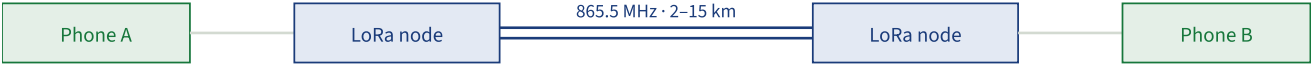


Figure 7.1 — The handset software is identical in both topologies. Only what sits at the far end of the Bluetooth connection differs.

SECTION 08

Evaluation methodology

Eighty per cent of the assessment is measurable, so the application is instrumented from the first commit and prints its own scorecard.

The stated criteria are **Accuracy 40 %**, **Latency 20 %** and **Efficiency 20 %**. Between them they force a real harness over held-out data under noise, instrumentation shipped inside the application rather than a stopwatch, and quantisation behind a tiered wake-up path.

Measurement conditions

Figures are reportable only from the entry-tier target handset, in a release build, after a **thirty-minute thermal soak**, over at least a hundred utterances, as median and 95th percentile. Entry-tier handsets throttle after about ten minutes of continuous inference and real-time factor can double, so sustained numbers are both more honest and more defensible than peak ones.

Accuracy — reported at four noise levels

Word error rate is measured on held-out benchmark data — IndicSUPERB, Kathbath, IndicVoices, FLEURS — never on self-recorded samples. Clean-room figures do not represent a flood embankment.

CONDITION	TARGET	WERCOMMENT
Clean, read speech	< 12	%Reference figure
+20 dB SNR	< 15	%Quiet room, realistic
+10 dB SNR	< 22	%Vehicle, crowd
+5 dB SNR	< 32	%Adverse; template fallback engages
Critical vocabulary, biased	< 6	%The figure that matters operationally

The last row is the important one: overall word error rate is a poor proxy for usefulness, and what matters is whether मदद, आग, sector numbers and callsigns survive. It is reported with and without biasing in the same table, so the delta is attributable to a deliberate engineering decision.

Latency and efficiency targets

LATENCY	TARGET	EFFICIENCY	TARGET
End to end, push-to-talk	500–800 ms	Installer, two languages	< 30 MB
End to end, phone mode	750–1100 ms	Per additional language	< 70 MB
First synthesis chunk	150–250 ms	Resident memory	< 350 MB
Real-time factor, ASR	< 0.30	Idle CPU while listening	< 2 %
Real-time factor, TTS	< 0.25	Standby endurance	> 8 h

Output artefacts

FILE	CONTENTS, EVERY ROW NAMING ITS DEVICE AND BUILD
scorecard.csv	Per language: WER at four SNRs, critical-term error with and without biasing, MOS, model sizes, RTF
latency.csv	Per utterance: every stage boundary, transport, payload size, end-to-end delta
resource.csv	Sampled CPU, resident memory, battery drain, thermal state

These three files convert the submission from a set of claims into a set of results.

SECTION 09

Risk and compliance

Twenty-nine identified failure modes, each with a designed response and a single owner. Severity reflects impact on the assessed criteria, not merely on the software.

ID	RISK	SEV	MITIGATION
T-01	Word error rate collapses in real acoustic noise	High	RNNoise pre-processing; evaluate and report at four SNRs rather than clean only
T-02	Digits and callsigns misrecognised — the tokens that carry meaning	High	Contextual biasing; separate critical-term metric tracked throughout
T-03	Thermal throttling doubles real-time factor mid-demonstration	High	Thirty-minute soak testing from week 6; sustained figures reported
T-05	No adequate open synthesis model exists for Odia	Med	Identified as the expected gap; Coqui VITS trained on IIT Madras data
T-08	Stream framing defect corrupts frames under load	Med	Mandatory read loop, CRC, magic-byte resynchronisation, decoder fuzzing
S-01	Fraudulent alert injected at maximum volume	High	AES-256-GCM; the sender identity sits inside the authenticated region
S-03	Recognition error inverts meaning — "do not evacuate"	High	Confidence on the wire; confirmation required for alerts; negations weighted
P-01	Model work consumes the schedule, loop unproven late	High	The loop closes in week 3 on the weakest acceptable models
P-02	Development on flagship hardware conceals failures	High	Entry-tier handset acquired in week 1; every benchmark taken on it

Open-source compliance

The problem statement prohibits proprietary, closed-source and commercial voice-activation SDKs. No such component appears anywhere in the system; every model executes on the device and no network request is made at runtime. The claim is enforced by the platform rather than asserted by us — the shipped manifest declares no **INTERNET** permission.

COMPONENT	LICENCE	ROLE
sherpa-onnx	Apache-2.0	Recognition, synthesis, VAD and endpointing
ONNX Runtime	MIT	Inference engine, XNNPACK on ARM
AI4Bharat IndicConformer	Permissive	Acoustic models, all ten languages
Piper voices	MIT	Primary synthesis voices
Silero VAD	MIT	Tier-1 voice activity detection
espeak-ng	GPL-3.0 ⚠	Phonemisation — copyleft, disclosed
Meta MMS	CC-BY-NC ⚠	Gap-filler only — non-commercial, disclosed

Two components carry restrictive licences. Both are declared here rather than left to be discovered, and any language pack containing a non-commercial model is marked as such in the application itself.

SECTION 10

Status, plan and conclusion

Where the project stands

The system is fully specified — roughly thirty-seven thousand words covering requirements traceability, architecture, a byte-level wire protocol, both speech engines, transport, security, evaluation methodology, a risk register and the complete interface design. Implementation has not begun; the task breakdown contains 207 items across eight weeks.

WEEK	OBJECTIVE	GATE
1	Capture and recognise	Hindi speech produces text on the target handset, offline
2	Transport	Typed text crosses two phones; decoder survives fuzzing
3	Close the loop	Speech in on A, speech out on B
4	Real models	Five languages; the harness produces a scorecard
5	Modes and alerts	An alert wakes a locked, silenced handset
6	Latency and reach	Encryption, four transports, first thermal soak
7	Coverage	Ten languages; field and relay soaks clean
8	Evidence	Full scorecard; three clean rehearsals

THE GOVERNING RULE

Week three is the hinge. A working loop with mediocre models can be improved under any amount of time pressure; excellent models with no loop cannot be demonstrated at all. Model quality is an upgrade path, never a prerequisite.

Conclusion

iTantra is not a speech-recognition demonstration with a Bluetooth link attached. It is a communications system that uses speech models as its codec. That framing matches the problem statement's own language — *neural transceiver radio access for low bitrate links* — and it is the framing under which every decision in this report is coherent rather than arbitrary.

The measure of success is narrow and testable: two handsets in aeroplane mode, ten languages, sub-second delivery, an installer under thirty megabytes, idle CPU below two per cent verified by trace rather than assertion, word error rate reported per language and per noise condition, alerts that wake a locked and silenced handset — and the same bytes crossing a real radio link.

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